

- ~~Caesar~~ Caesar cipher —
- shift cipher
- monoalphabetic substitution cipher

Encryption: $CT = (PT + k) \bmod 26$

Decryption: $PT = (CT - k) \bmod 26$

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$$\begin{aligned} D &= 23 \\ A &= 0 \\ T &= 19 \\ W &= 22 \end{aligned}$$

~~CT = WTAAB~~, $k = 15$

$$\begin{array}{ccccccc} \downarrow & \downarrow & \downarrow & \downarrow & \searrow & & \\ \underline{7} & \underline{4} & \underline{25} & \underline{14} & \underline{19} & & \\ H & E & L & L & O & & \end{array}$$

$4 - 15 = -11 = 25 - 11 = 14$

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PT = ~~61~~ ~~we~~ ~~ried~~ ~~AK~~ ~~Neon~~
 K = 3
 25 7 20 11 6 7 3 22 6 17 17 6
 CT = ZH ULGH DW QRRQ

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CT = ~~65~~ ~~GZD~~ ~~KWK~~ ~~YDX~~ ~~MLW~~ ~~JXA~~
 Key k = 5
 1 20 -1 = 24 5 19 24 18 70 17 4 18 24
 PT = ~~BUY~~ FIF TYS HAR ESC

Playfair cipher →

— wheatstone cipher

— manual symmetric encryption technique

— 5x5 matrix constructed using keyword

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letter → row

other letter → col.

K = MONARCHY

Matrix

PT: A | T | A | K
 CT: S | R | S | E

~~same row~~ ~~same col~~

PT: A | T | A | K

CT: R | S | S | DE

same row → shift right

~~PT: J | R | S | R | U | ME | NT | EX~~

~~CT: G | A | T | S | M | Z | ME | RQ | EX~~

same col →
move down

PT: J | R | S | R | U | ME | NT | EX

CT: G | A | T | S | M | Z | CL | RQ | XA

PT: B | A | K | X | EO | DX

CT: I | B | S | U | PM | NA

PT: M | O | S | Q | UE

CT: O | N | R | S | ML

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$K = \text{ENCRYPT}$

PT: ~~THEY~~ ~~HIM~~ ABOUT ME

~~T E H L P H I D M A~~
~~P N I~~
~~B O U T M E~~

Matrix

E	N	C	R	Y
P	T	A	B	D
F	G	H	I	K
L	M	O	Q	S
U	V	W	X	Z

~~TE~~ | ~~EX~~ | ~~WH~~ | ~~IM~~ | ~~AB~~ | ~~OU~~ | ~~TH~~ | ~~EX~~

CT: PN | QV | OF | GQ | BD | LW | QV | RU

~~EX~~ | ~~AS~~ | ~~EX~~ | ~~CA~~ | ~~EX~~ | ~~EP~~ | ~~HE~~ | ~~RS~~

CT: EO | DO | QK | ~~AM~~ | OE | PB | FC | YQ

— Hill cipher —

— polygraphic substitution ciphers

— works on multiple letters at the same time

$$CT = E(PT, K) = (P \times K) \bmod 26$$

$$PT = D(K, CT) = (C K^{-1}) \bmod 26 = (P \times K \times K^{-1}) \bmod 26$$

encryption

~~K₁ P₁ P₂ P₃~~

$$(C_1 \ C_2 \ C_3) = (P_1 \ P_2 \ P_3) \begin{pmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{pmatrix} \bmod 26$$

1x1 1x3 3x3

decryption

~~K₁ P₁ P₂ P₃~~

$$(P_1 \ P_2 \ P_3) = (C_1 \ C_2 \ C_3) \begin{pmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{pmatrix}^{-1} \bmod 26$$

3x1 3x3 3x3

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$$P_T = \text{'abc'}$$

$$K = \begin{bmatrix} 3 & 0 & 2 \\ 2 & 0 & -2 \\ 0 & 1 & 1 \end{bmatrix}$$

$$C_T = \begin{bmatrix} 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 0 & 2 \\ 2 & 0 & -2 \\ 0 & 1 & 1 \end{bmatrix} \pmod{26}$$

$$C_T = \begin{bmatrix} 2 & 2 & 0 \end{bmatrix} \pmod{26}$$

$$C_T = \text{'ccca'}$$

$$P_T = \begin{bmatrix} 2 & 2 & 0 \end{bmatrix} \begin{bmatrix} 0.2 & 0.2 & 0 \\ -0.2 & 0.3 & 1 \\ 0.2 & -0.3 & 0 \end{bmatrix} \pmod{26}$$

$$P_T = \begin{bmatrix} \overline{a} & \overline{b} & \overline{c} \\ 0 & 1 & 2 \end{bmatrix}$$

other method OR

→ Rail Fence

→ Single Columnar (Block)

→ Double Columnar

→ Rail Fence cipher →

→ zigzag cipher

→ write on alternate lines & read

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PT: "This is a secret message"

K: 2

T I I A E R T E S G
H S S S C E M S A E

CT: T I I A E R T E S G H S S S C E M S A E

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PT: "Defend the east wall?"

K: 3

D N E T L E E D H E S W L X F T A A X

CT: D N E T L E E D H E S W L X F T A A X

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K: ZEBRAS

PT: ~~C W~~ ~~are~~ ~~dy~~ ~~are~~ ~~are~~ ~~are~~ ~~are~~

6	3	2	4	1	5
Z	E	B	R	A	S
W	E	A	R	E	D
I	S	C	O	V	E
R	E	D	F	L	E
E	A	T	O	N	C
E	X	X	X	X	X
	↓	↓	↓	↓	↓
	Q	K	I	U	Y

CT: ~~EVLNU~~ ~~ACDTK~~ ~~SEAB~~ ~~ROFOJ~~
~~DEECY~~ ~~WIREG~~

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K: BOAT

PT: ~~'geekforgeeks'~~

2	3	1	4
B	O	A	T
G	E	E	K
S	F	O	R
G	E	E	K
S	X	X	X

CT: ~~EDDXX~~ ~~GGSGS~~ ~~EEEX~~ ~~ERRKX~~

decryption

- col len, → msg. len ÷ keylen, colwise
- write msg, in alphabetical order, then reorder columns

CT: ~~6TRXEEYXGEXGCHXANXNXXEYXN~~
 Key: ~~analyst~~

~~AALNSTY~~
~~TRXEE~~

A	A	L	N	S	T	Y
T	R	L	E	E	L	I
G	C	I	G	E	H	A
V	A	N	T	N	C	T
E	C	Y	E	N	E	N

A	A	L	N	S	T	Y
T	E	C	H	N	T	E
R	E	I	A	T	E	N
L	I	G	L	N	C	E
E	G	E	A	C	Y	N

A	N	A	L	Y	S	T
T	H	E	C	E	A	T
R	A	L	I	N	T	E
L	L	I	G	E	N	C
E	A	G	E	N	C	Y

PT: THE CENTRAL INTELLIGENCE AGENCY

- Double transposition cipher —
- two app. of columnar transposition
 - might be same / diff. key

PT: "We are disappointed free of cost"

K₁: "Zebras"

K₂: "help"

2 6	3	2	4	1	5
Z	E	B	R	A	S
W	E	A	R	E	D
I	S	C	O	V	E
R	E	D	F	L	E
E	A	T	O	N	C
E	X	X	X	X	X
	↓	↓	↓	↓	↓
Q	K	J	U	Y	

CT₁: ~~EXVNU ACDTK ESEDA ROPOS DECK WDEA~~

2	1	3	4
H	E	L	P
E	V	L	N
U	A	C	D
T	K	E	S
E	A	Q	R
D	F	O	J
D	E	E	C
Y	W	I	R
E	E	X	X
		↓	↓
		K	G

CT₂: ~~QVAKAFWE~~ ~~W~~ EUTEDDYE
LC E Q O E J K N D S R J C R G

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$\mathbb{Z}_4$

$$(17 + 27) \bmod 14$$

$$44 \bmod 14 = \underline{\underline{2}}$$

$$14 \times 3 = 42$$

$\mathbb{Z}_{13}$

$$(12 - 43) \bmod 13$$

$$-31 \bmod 13 = -5$$

$$13 \times 2 = 26$$

$$13 - 5 = \underline{\underline{8}}$$

$\mathbb{Z}_{19}$

$$(123 \times -10) \bmod 19$$

$$-1230 \bmod 19 = -14$$

$$+19 - 14 = 5$$

$$10^n \bmod x = (10 \bmod x)^n \bmod x$$

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$$\begin{aligned} 10^2 \bmod 3 &= (10 \bmod 3)^2 \bmod 3 \\ &= \cancel{100 \bmod 3} 4 \bmod 3 = 1 \end{aligned}$$

$$\begin{aligned} 10^3 \bmod 7 &= (10 \bmod 7)^3 \bmod 7 \\ &= 27 \bmod 7 = 6 \end{aligned}$$

$$\begin{aligned} 10^4 \bmod 6 &= (10 \bmod 6)^4 \bmod 6 \\ &= 256 \bmod 6 = 4 \end{aligned}$$

→ Additive Inverse $a + b \equiv 0 \pmod{n}$

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~~10 - 6 = 4~~ Additive Inverse of 6 in $\mathbb{Z}_{10}$

$$10 - 6 = 4$$

All additive inverse pairs in $\mathbb{Z}_{10}$

$(0, 10), (1, 9), (2, 8), (3, 7), (4, 6), (5, 5)$

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multiplicative inverses in $\mathbb{Z}_{10}$ — anything ending with 1

$$(1, 1) = 1 \mod 10 = 1$$

2, x

$$(3, 7) = 21 \mod 10 = 1$$

~~4, x~~ ~~6, x~~ ~~8, x~~ ~~10, x~~

$$(9, 9) = 81 \mod 10 = 1$$

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multiplicative inverses in $\mathbb{Z}_{11}$

$$(1, 1) = 1 \mod 11 = 1$$

$$(2, 6) = 12 \mod 11 = 1$$

$$(3, 4) = 12 \mod 11 = 1$$

$$(5, 9) = 45 \mod 11 = 1$$

$$(7, 8) = 56 \mod 11 = 1$$

$$(9, 5) = 45 \mod 11 = 1$$

— Extended Euclidean algo, —

— Int, a in $\mathbb{Z}_n$ has multiplicative inverse iff $\gcd(n, a) = 1 \pmod{n}$

— Bezout Identity — assume a, b are integers, there exists x & y such that:
$$ax + by = \gcd(a, b)$$

— utilizes Bezout's Identity & Euclid's algo.

— $\gcd(a, b) = 1$ — a, b are coprime

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$$\gcd(2740, 1760)$$

$$\begin{array}{r} 1760 \\ \times 2 \\ \hline 3520 \end{array}$$

q	r ₁	r ₂	r
1	2740	1760	980
1	1760	980	200
4	980	200	180
1	200	180	20
9	180	20	0
	<u>20</u>		

— Extended Euclidean algo, —

$$s \times a + t \times b = \gcd(a, b)$$

Init, $r_1 \leftarrow a, r_2 \leftarrow b, s_1 \leftarrow 1, s_2 \leftarrow 0, t_1 \leftarrow 0, t_2 \leftarrow 1$
Slide 224 $s \leftarrow s_1 - q \times s_2, s_1 \leftarrow s_2, s_2 \leftarrow s$
 same for t

$$a = 161$$

$$b = 28$$

~~Preceded~~

q	a	b	r	s ₁	s ₂	t ₁	t ₂
5	161	28	21	1	0	0	1
1	28	21	7	0	1	1	-5
3	21	7	0	1	-1	-5	6
	<u>7</u>				<u>-1</u>		<u>6</u>
	↓				↓		↓

gcd.

$$s = -1, t = 6$$

q	a	b	r	s ₁	s ₂	t ₁	t ₂
5	161	28	21	1	0	0	1
1	28	21	7	0	1	1	-5
3	21	7	0	1	-1	-5	6
	↓				↓		↓

gcd,

$$a = 26 \quad b = 11$$

$$t = t_1 - q t_2$$

q	a	b	r	t ₁	t ₂	t
2	26	11	4	0	1	-2
2	11	4	3	1	-2	5
1	4	3	1	-2	5	-7
3	3	1	0	5	-7	26
	1	0		-7	26	

↓
gcd

$$-7 \bmod 26 = 19$$

$$11 \times 19 \bmod 26 = 1$$

$$a = 100, \quad b = 23$$

$$t = t_1 - q t_2$$

q	a	b	r	t ₁	t ₂	t
4	100	23	8	0	1	-4
2	23	8	7	1	-4	9
1	8	7	1	-4	9	-13
7	7	1	0	9	-13	100
	1	0		-13	100	

$$-13 \bmod 100 = 87$$

$$23 \times 87 \bmod 100 = 1$$

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$$11^{43} \bmod 13$$

$$43 = 32 + 8 + 2 + 1$$

~~$$11 \bmod 13 = 11$$~~

~~$$11^2 \bmod 13 = 121 \bmod 13 = 4$$~~

~~$$11^4 \bmod 13 = 16 \bmod 13 = 3$$~~

~~$$11^8 \bmod 13 = 9 \bmod 13 = 9$$~~

~~$$11^{16} \bmod 13 = 81 \bmod 13 = 3$$~~

~~$$11^{32} \bmod 13 = 9 \bmod 13 = 9$$~~

$$11^{43} \bmod 13 = (9 \times 9 \times 4 \times 11) \bmod 13$$

~~$$11^{43} \bmod 13 = 81 \bmod 13 \times 44 \bmod 13$$~~

$$11^{43} \bmod 13 = (81 \times 44) \bmod 13$$

$$11^{43} \bmod 13 = 81 \bmod 13 \times 44 \bmod 13$$

$$11^{43} \bmod 13 = (5 \times 3) \bmod 13 = 2$$

— Fermat's Little Theorem —

— If p — prime no,

$$a^{p-1} \equiv 1 \pmod{p}$$

$$\Rightarrow a \cdot a^{p-2} \equiv 1 \pmod{p}$$

$$\Rightarrow a^{-1} \pmod{p} \equiv a^{p-2} \pmod{p}$$

$$a^{p-1} \equiv 1$$

$$a \cdot a^{p-2} \equiv 1$$

$$a^{p-2} \equiv a^{-1}$$

$$3^{12} \bmod 11 \quad 12 = 8 + 4$$

$$3^2 \bmod 11 = 9$$

$$3^4 \bmod 11 = 81 \bmod 11 = 4$$

$$3^8 \bmod 11 = 16 \bmod 11 = 5$$

$$3^{12} \bmod 11 = 20 \bmod 11 = 9$$

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~~$$8^{-1} \bmod 17 = 8^{15} \bmod 17$$~~

~~800~~

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$$8^{-1} \bmod 17 = 8^{15} \bmod 17$$

$$8 \bmod 17 = 8$$

$$8^2 \bmod 17 = 64 \bmod 17 = 13$$

$$8^4 \bmod 17 = 169 \bmod 17 = 16$$

$$8^8 \bmod 17 = 256 \bmod 17 = 1$$

$$8^{15} \bmod 17 = (8 \times 13 \times 16) \bmod 17$$

$$8^{-1} \bmod 17 = \cancel{8 \times 13 \times 16} 15 \bmod 17$$

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$$22^{-1} \bmod 211$$
$$= 22^{209} \bmod 211$$

$$209 = 128 + 64 + 16 + 1$$

$$19^{-1} \bmod 17$$

$$a=17 \quad b=19$$

$$t = t_1 - q t_2$$

$$q \quad a \quad b \quad r \quad t_1 \quad t_2 \quad t$$

$$0 \quad 17 \quad 19 \quad 17 \quad 0 \quad 1 \quad 0$$

$$1 \quad 19 \quad 17 \quad 2 \quad 1 \quad 0 \quad 1$$

$$8 \quad 17 \quad 2 \quad 1 \quad 0 \quad 1 \quad -8$$

$$2 \quad 2 \quad 10 \quad 1 \quad -8 \quad 9$$

$$10 \quad -8 \quad 9$$

$$-8 \bmod 17$$
$$= 9$$

$$22^{209} \bmod 211 = (22 \times 36 \times 56 \times 182) \bmod 211$$
$$22^{-1} \bmod 211 = 48$$

$$5^{-1} \bmod 23$$

$$= 5^{21} \bmod 23$$

$$21 = 16 + 4 + 1$$

$$5 \bmod 23 = 5$$

$$5^2 \bmod 23 = 2$$

$$5^4 \bmod 23 = 4$$

$$5^8 \bmod 23 = 16$$

$$5^{16} \bmod 23 = 3$$

$$5^{21} \bmod 23 = (5 \times 4 \times 3) \bmod 23 = 14$$

$$5^{-1} \bmod 23 = 14$$

$$60^{-1} \bmod 101$$
$$= 60^{99} \bmod 101$$

$$99 = 64 + 32 + 2 + 1$$

~~$$60 \bmod 101 = 60$$~~

~~$$60^2 \bmod 101 = 65$$~~

$$60^4 \bmod 101 = 84$$

$$60^8 \bmod 101 = 87$$

$$60^{16} \bmod 101 = 95$$

~~$$60^{32} \bmod 101 = 36$$~~

~~$$60^{64} \bmod 101 = 84$$~~

$$60^{99} \bmod 101 = (60 \times 65 \times 36 \times 84) \bmod 101$$

$$60^{-1} \bmod 101 = \underline{\underline{32}}$$

— correctness —

$$\text{Dec}_k(\text{Enc}_k(m)) = k \oplus (k \oplus m) \\ = (k \oplus k) \oplus m = m$$

— key chosen randomly

PT: $x = (x_1, x_2, \dots, x_n)$

CT: $y = (y_1, y_2, \dots, y_n)$

Key: $k = (k_1, k_2, \dots, k_n)$

$$e_k(x) = (k_1 + x_1, k_2 + x_2, \dots, k_n + x_n) \bmod m \\ d_k(y) = (y_1 - k_1, y_2 - k_2, \dots, y_n - k_n) \bmod m$$

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msg: 'IF'

ASCII code: (1001001 1000110)

key: (1010110 0110001)

enc

PT: 1001001 1000110

k: $\oplus$ 1010110 0110001

CT: 0011111 110111 ✓

dec

CT: 0011111 110111

k: $\oplus$ 1010110 0110001

PT: 1001001 1000110 ✓

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$$GF(2^3) = GF(8)$$

$$(a_2, a_1, a_0)$$

$$(0, 0, 0) = 0$$

$$(0, 0, 1) = 1$$

$$(0, 1, 0) = a$$

$$(0, 1, 1) = a+1$$

$$(1, 0, 0) = a^2$$

$$(1, 0, 1) = a^2+1$$

$$(1, 1, 0) = a^2+a$$

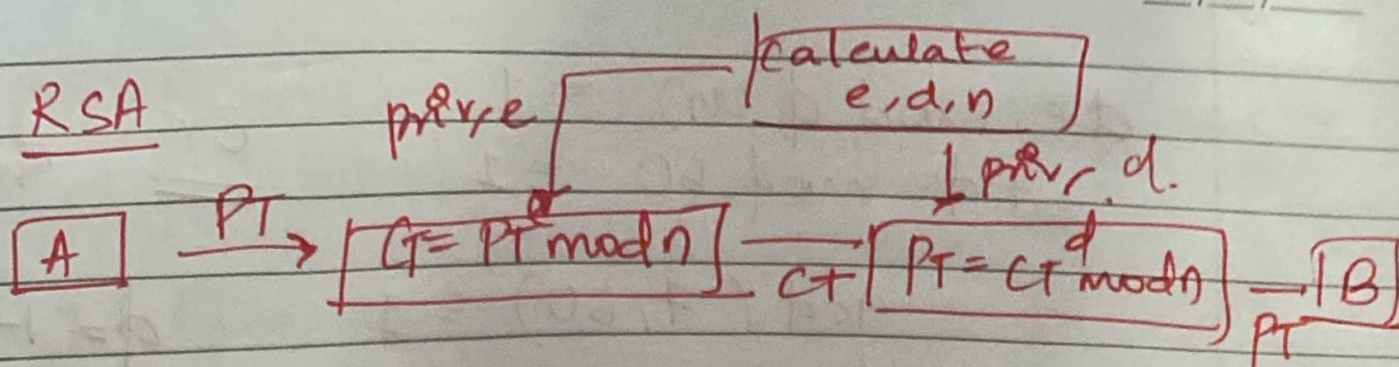
$$(1, 1, 1) = a^2+a+1$$

$$GF(2^3) = \{0, 1, a, a+1, a^2, a^2+1, a^2+a, a^2+a+1\}$$

— poly $f(x)$ over field F irreducible if $f(x)$ can't be expressed as prod of 2 poly, both over F & degree lower than degree of $f(x)$

Prime polynomial

RSA



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enc $PT = 9$

$p = 7, q = 11$

$n = p \times q = 77$

$\phi(n) = (p-1) \times (q-1) = 60$

$e \rightarrow$ co prime with 60

$e \rightarrow \underline{7}$

public key $(n, e) = (77, 7)$.

enc

$$CT = PT^e \bmod n$$

$$CT = 9^7 \bmod 77$$

$$CT = 37$$

$$de \bmod \phi(n) = 1$$

$$7d \bmod 60 = 1$$

or ~~de mod 60 = 1~~

$$\gcd(7, 60) = 1$$

$$d = \frac{1 + k\phi(n)}{e}$$

use k till divisible

$$60 \bmod 7 = 4$$

$$7 \bmod 4 = 3$$

$$4 \bmod 3 = 1$$

$$3 \bmod 1 = 0$$

$$d = \frac{1 + 0}{7}$$

$$d = \frac{1 + 60}{7}$$

$$1 = 4 - 3$$

$$1 = 4 - (7 - 4)$$

$$1 = 2(4) - 7$$

$$1 = 2(60 - 8(7)) - 7$$

$$1 = 2(60) - (17)(7)$$

$$d = \frac{181}{7}$$

$$d = \frac{241}{7}$$

$$d = \frac{301}{7}$$

$$d = 43$$

2.

$$-17 \bmod 60 = 43$$

$$d = 43$$

dec

$$PT = 37^{43} \bmod 77$$

$$43 = 32 + 8 + 2 + 1$$

$$37 \bmod 77 = 37$$

$$37^2 \bmod 77 = 60$$

$$37^4 \bmod 77 = 58$$

$$37^8 \bmod 77 = 53$$

$$37^{16} \bmod 77 = 37$$

$$37^{32} \bmod 77 = 60$$

$$PT = (37 \times 60 \times 53 \times 60) \bmod 77$$

$$PT = 9$$

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1. $p = 13$
 $q = 17$

$$n = 13 \times 17 = 221$$

$$\phi(n) = 12 \times 16 = 192$$

given $e = 35$

public key $(221, 35)$

$$d = \frac{1 + k\phi(n)}{e}$$

$$d = \frac{1 + 192 \times 35}{35}$$

$$d = \frac{1 + 384}{35}$$

$d = 11$. private key $(221, 11)$

2. $p = 3$
 $q = 13$

$$n = 39$$

$$\phi(n) = 24$$

~~$e = 7$~~

$$e = \frac{1 + k\phi(n)}{d}$$

$$e = \frac{1 + 24 \times 7}{7}$$

$$e = \frac{1 + 48}{7} = 7$$

$$CT = P^e \bmod n$$

$$CT = 5^7 \bmod 39$$

$$\begin{aligned} p &= 3 \\ q &= 13 \\ e &= 3 \\ d &= 7 \end{aligned}$$

$$n = 39, \phi(n) = 24$$

$$d = \frac{1 + k\phi(n)}{e}$$

$$d = \frac{1 + 24}{3} = 8$$

$$P_T = 5$$

$$CT = P_T^e \bmod n$$

$$CT = 5^3 \bmod 39$$

$$CT = 125 \bmod 39 = \underline{8}$$

$$P_T = CT^d \bmod n$$

$$P_T = 8^7 \bmod 39$$

$$P_T = \underline{5}$$

$$\begin{aligned} 3. \quad p &= 3 \\ q &= 11 \\ n &= 33 \\ \phi(n) &= 20 \\ d &= 7 \\ P_T &= 5 \end{aligned}$$

$$e = \frac{1 + k\phi(n)}{d}$$

$$e = \frac{1 + 20}{7} = \underline{3}$$

$$CT = P_T^e \bmod n$$

$$CT = 5^3 \bmod 33 = 26$$

- Square & Multiply algo, —
- scanning ~~bit from~~ of exp, from left (MSB) to right (LSB)
 - every iter, — current result squared
 - if scanned exp, bit = 1, multiplication of current result by x following the squaring

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$$x^{26} = x^{11010_2} = x^{(\overbrace{h_4 h_3 h_2 h_1 h_0}^{1 \ 1 \ 0 \ 1 \ 0})_2}$$

start with h_4

#0 $x = x^1$ $h_4 = 1$

#1a $(x^1)^2 = x^2 = x^{10_2}$

#1b $x^2 \cdot x = x^3 = x^{11_2}$ since $h_3 = 1$

#2 $(x^3)^2 = x^6 = x^{110_2}$ $h_2 = 0$

#3a $(x^6)^2 = x^{12} = x^{1100_2}$

#3b $(x^{12}) \cdot x = x^{13} = x^{1101_2}$ $h_1 = 1$

#4 $(x^{13})^2 = x^{26} = x^{11010_2}$, $h_0 = 0$

— Algo, —

x/p — base elem x

exp, $H = \sum_{i=0}^t h_i 2^i$ where $h_i \in \{0, 1\}$
 $\sum_{i=0}^t h_i = 1$

mod n

O/P $\rightarrow x^H \text{ mod } n$

init, $r = x$

For $i = t-1$ down to 0

$r = r^2 \text{ mod } n$

If $h_i = 1$

$r = r \cdot x \text{ mod } n$

Return (r)

— complexity of square & multiply algo, ^{exp. H} bit length $t+1$
 $\log_2 H = t+1$

H — hamming weight — no. of 1s in binary rep
↓
no. of sq, & mul, — no. of mul ~~equal~~ equal to H

$$\# \text{ SQ} = t$$

$$\# \text{ MUL} = 0.5t$$

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$$\text{normal exp} = 2^{1024} \approx 10^{300}$$

$$\text{sq / \& mul} = 1.5 \times 1024 = 1536$$

diff. in algo, with linear complexity &
 $\log_2$ complexity

PCA comp. all

— Diffie-Hellman Key Exchange Algo, —

— Algorithmically — easy to multiply two large primes but diff, to recover these primes from their product.

— Perform ops, that sender & receiver comm, via public channel & sharing secret key can reverse but, eavesdropper cannot

— PKC —

— computationally infeasible to det, dec, key given only cryptos algo & enc, key

— Algo, —

Step 1: q — prime no, & α — primitive root of q

Step 2: User A key gen, $X_A < q$

— Public key A $Y_A = \alpha^{X_A} \bmod q$

— Y_A shared with user B

Step 3: User B key gen, $X_B < q$

— Public key B $Y_B = \alpha^{X_B} \bmod q$

— Y_B shared with user A

Step 4: secret key by user A: $K_A = (Y_B)^{X_A} \bmod q$

Step 5: " B: $K_B = (Y_A)^{X_B} \bmod q$

Algebraically can be shown that
 $K_A = K_B$

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$$n = 11 \quad (q)$$

$$g = 7 \quad (x)$$

$$x = 3 \quad (y_A), \quad y = 5 \quad (y_A).$$

~~$x_A = 7$~~

$$y_A = 7^3 \bmod 11 = \underline{2}$$

$$y_B = 7^5 \bmod 11 = \underline{10}$$

$$K_A = (y_B)^{x_A} \bmod q$$

$$K_A = 10^3 \bmod 11 = \underline{10}$$

$$K_B = (y_A)^{x_B} \bmod q$$

$$K_B = (2)^5 \bmod 11 = \underline{10}$$

$$K_A = K_B$$

$$z = 4 \quad (x_C)$$

$$y_C = 7^4 \bmod 11 = \underline{3}$$

~~$$K_{CA} = (y_A)^{x_C} \bmod q$$~~

~~$$K_{CA} = 2^4 \bmod 11 = 8$$~~

$$K_{CA} = (y_B)^{x_C} \bmod q$$

$$K_{CA} = 10^4 \bmod 11 = 1$$

$$K_{CB} = (y_A)^{x_C} \bmod q$$

$$K_{CB} = 2^4 \bmod 11 = 5$$

Slide 91

$$q = 23$$

$$\alpha = 9$$

$$x_A = 4$$

$$x_B = 3$$

$$y_A = \alpha^{x_A} \bmod q = 9^4 \bmod 23 = 6$$

$$y_B = \alpha^{x_B} \bmod q = 9^3 \bmod 23 = \underline{16}$$

$$K_A = (y_B)^{x_A} \bmod q = (16)^4 \bmod 23 = \underline{9}$$

$$K_B = (y_A)^{x_B} \bmod q = (6)^3 \bmod 23 = \underline{9}$$

$$K_A = K_B = \underline{9} \text{ — shared secret}$$

Slide 92

$$q = 353$$

$$\alpha = 3$$

$$x_A = 97$$

$$x_B = 233$$

$$y_A = \alpha^{x_A} \bmod q = 3^{97} \bmod 353$$

$97 = 64 + 32 + 1$

$$\cancel{y_A} \quad 3 \bmod 353 = 3$$

$$3^2 \bmod 353 = 9$$

$$3^4 \bmod 353 = 81$$

$$3^8 \bmod 353 = 207$$

$$3^{16} \bmod 353 = 156$$

$$\checkmark 3^{32} \bmod 353 = 140$$

$$\checkmark 3^{64} \bmod 353 = 185$$

$$y_A = 3^{97} \bmod 353 = (3 \times 140 \times 185) \bmod 353 = \underline{40}$$

$$Y_B = \alpha^{x_B} \bmod q = 3^{233} \bmod 353$$

$$233 = 128 + 64 + 32 + 8 + 1$$

$$\cancel{3} \bmod 353 = 3$$

$$\cancel{3^8} \bmod 353 = 207$$

$$\cancel{3^{32}} \bmod 353 = 140$$

$$\cancel{3^{64}} \bmod 353 = 185$$

$$\cancel{3^{128}} \bmod 353 = 337$$

(Y_A from calculations)

$$Y_B = 3^{233} \bmod 353 = (3 \times 207 \times 140 \times 185 \times 337) \bmod 353$$

$$Y_B = 248$$

$$K_A = (Y_B)^{x_A} \bmod q = (248)^{97} \bmod 353$$

$$97 = 64 + 32 + 1$$

$$\cancel{248} \bmod 353 = 248$$

$$248^2 \bmod 353 = 82$$

$$248^4 \bmod 353 = 17$$

$$248^8 \bmod 353 = 289$$

$$248^{16} \bmod 353 = 213$$

$$\cancel{248^{32}} \bmod 353 = 185$$

$$\cancel{248^{64}} \bmod 353 = 337$$

$$K_A = (248)^{97} \bmod 353 = (248 \times 185 \times 337) \bmod 353$$

$$= 160$$

$$K_B = (Y_A)^{x_B} \bmod q = 40^{233} \bmod 353$$

$$2^{23} = 128 + 64 + 32 + 8 + 1$$

$$40 \bmod 353 = 40$$

$$40^2 \bmod 353 = 188$$

$$40^4 \bmod 353 = 44$$

$$40^8 \bmod 353 = 171$$

$$40^{16} \bmod 353 = 298$$

$$40^{32} \bmod 353 = 187$$

$$40^{64} \bmod 353 = 22$$

$$40^{128} \bmod 353 = 131$$

$$K_B = 40^{235} \bmod 353 = (40 \times 171 \times 187 \times 22 \times 131) \bmod 353 = \underline{\underline{160}}$$

$$K_A = K_B = \underline{\underline{160}}$$

Slide 94

$$q = 23$$

$$\alpha = 5$$

$$x_A = 6$$

$$x_B = 15$$

$$y_A = \alpha^{x_A} \bmod q = 5^6 \bmod 23 = 8$$

$$y_B = \alpha^{x_B} \bmod q = 5^{15} \bmod 23 =$$

$$15 = 8 + 4 + 2 + 1$$

$$5^8 \bmod 23 = 5$$

$$5^2 \bmod 23 = 2$$

$$5^4 \bmod 23 = 4$$

$$5^8 \bmod 23 = 16$$

$$y_B = 5^{15} \bmod 23 = (5^8 \times 5^4 \times 5^2 \times 5^1) \bmod 23 = 19$$

//_

$$K_A = (Y_B)^{X_A} \bmod q = 19^6 \bmod 23 = \underline{\underline{2}}$$

$$K_B = (Y_A)^{X_B} \bmod q = 8^{15} \bmod 23$$

$$\cancel{8} \bmod 23 = 8$$

$$\cancel{8^2} \bmod 23 = 18$$

$$\cancel{8^4} \bmod 23 = 2$$

$$\cancel{8^8} \bmod 23 = 4$$

$$\begin{aligned} K_B &= 8^{15} \bmod 23 = (8 \times 18 \times 2 \times 4) \bmod 23 \\ &= \underline{\underline{2}} \end{aligned}$$

$$K_A = K_B = 2$$

Slide 119

key gen,

$$p=29 \quad g=2 \quad b=5$$

$$B = g^b \text{ mod } p$$

$$B = 2^5 \text{ mod } 29 = \underline{\underline{3}}$$

enc,

$$PT=6$$

$$a=4$$

$$S = B^a \text{ mod } p$$

$$S = 3^4 \text{ mod } 29 = \underline{\underline{23}}$$

~~$23 \text{ mod } 29$~~
 ~~$a=29, b=23$~~
 ~~$a \times b \times \dots \times CT$~~
 $CT = PT \times S \text{ mod } p = 6 \times 23 \text{ mod } 29 = \underline{\underline{22}}$

$$A = g^a = \underline{\underline{16}}$$

dec.

$$S = A^b \text{ mod } p = 16^5 \text{ mod } 29 = \underline{\underline{23}}$$

$$S^{-1} \text{ mod } p = 23^{-1} \text{ mod } 29 = \underline{\underline{24}}$$

$$PT = CT \times S^{-1} \text{ mod } p$$

$$= 22 \times 24 \text{ mod } 29 = \underline{\underline{26}}$$

— Birthday attack — Brute force

— Derivation —

— assume non-leap year

— assume person eq. likely to be born on any day

Abuse
among

h/w parties, $P(2 \text{ ppl same}) = 1 - P(2 \text{ ppl diff})$

Compromise

Crypto, sys

$$= 1 - P(365/365) \times (364/365)$$

$$= 1 - 364/365$$

$$= 1/365$$

$P(n \text{ ppl with diff. bdays})$

$$= \left(\frac{365}{365}\right) \times \left(\frac{364}{365}\right) \times \dots$$

$$= \frac{365!}{(365-n)! \times 365^n}$$